

# Jacob Davis

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## Education

### Clemson University

Clemson, SC

*Bachelor of Science in Computer Science and Mathematical Sciences (GPA: 3.97)*

*Aug. 23 – May. 27*

- **Clubs & Affiliations:** Teaching Assistant (Data Structures and Algorithms), Honors Peer Mentor, Symphony Orchestra (Leader)
- **Relevant Coursework:** Operating Systems, Network Programming, Parallel Architecture, Quantum Computing, Databases
- **Technical Skills:** C++, Python, C, JavaScript, Go, SQL, Kubernetes, Docker, CUDA, JAX, PyTorch, OpenMP, OpenCL, Linux, Git

## Experience

### Google

Sunnyvale, CA

*Software Engineer Intern*

*May. 26 – Aug. 26*

- **Doubled** theoretical VM capacity per host and conserved up to **50%** of the host's buffer resources by engineering an on-demand allocation optimization in C++ on SmartNICs.
- **Mitigated a high-priority production alert** caused by buffer exhaustion across high-density machines by developing a thread-safe and **low-latency** asynchronous callback to dynamically expand packet processing queues.
- Prevented resource leakage for feature downgrades through implementing safe state restoration and safeguard clamping during **hitless transfers** within a cloud distributed system.

### Clemson Research Computing and Data

Clemson, SC

*Software Engineer Intern*

*Jan. 26 – May. 26*

- Decreased website latency by **20%** by optimizing an OpenAI LLM service with local caching in **Go** to lower **MySQL** database reads.
- Enabled LLM performance evaluation under **hundreds** of configs in a **Kubernetes** and **Slurm** cluster by building a benchmark suite.

### NASA Goddard Space Flight Center

Greenbelt, MD

*Software Engineer Intern*

*Jun. 25 – Jan. 26*

- Achieved a **5x performance increase** in a sky localization algorithm for compact binary mergers compared to **C** and **OpenMP** code on the CPU using **Python** and **JAX** to restructure the computation for the GPU.
- Reduced runtime by **99%** compared to sequential code by migrating open-source software used at gravitational-wave observatories to parallel processing frameworks using **CUDA**, **OpenCL**, and **OpenMP**.
- Presented at the 247th American Astronomical Society Meeting and won the “**Most Impactful Award**” at Clemson's HPC Day.

### Clemson University International Center for Automotive Research

Greenville, SC

*Software Engineer Intern*

*Aug. 24 – May. 25*

- Achieved **95% accuracy** in differentiating trails and vegetation using a custom dataset of outdoor environments by training and fine-tuning LiDAR and image classification machine learning models in **Python** and **PyTorch**.
- Deployed data collection for an autonomous vehicle by developing a sensor data system, working with Applied Research Associates.

### Cadence Design Systems

San Jose, CA

*Application Engineer Intern*

*May. 24 – Aug. 24*

- Pinpointed over **20 bugs** when testing the RTL code of a controller across **30+ features**, achieving **99%** functional coverage using **Universal Verification Methodology (UVM)**, **SystemVerilog**, Cadence Xcelium, and SimVision.
- Increased workflow efficiency by **50%** by using **shell scripting** and **C++** to automate simulation analysis in a **Linux** environment.

## Projects

### High-Performance Cluster Computing

*Jan. 25 – Present*

- Reached a score of **~30 GFLOPS** on a custom four-node **Raspberry Pi** mini-cluster using **Spack**, **Docker**, and **OpenMPI** to simulate Clemson's **TOP500** Palmetto Cluster.
- Won **first place** in the interviews for a climate emulator at SC25's **international** Student Cluster Competition (SCC).
- Prepare a six-undergraduate team for SC26's SCC by directing weekly training sessions and system administration as **team lead**.

### HPC Cluster System Admin Dashboard

*Jan. 26*

- Won the “**Best Use of MongoDB Award**” at HoyaHacks against **400+ participants** by engineering a real-time file monitoring system for HPC clusters featuring an interactive dashboard and terminal navigator using **JavaScript** and **Vue**.

### Computer Vision Paper Synthesizer

*Sep. 25*

- Won the “**Best Musical Hack Award**” at HackMIT against **300+ teams**, achieving **90% key detection accuracy** to play a synthesizer on paper using **Python**, **YOLOv8**, and **MediaPipe**.

### Asteroid Collision Simulator

*Sep. 25*

- Won the “**Most Inspirational Award**” at the NASA Space Apps Challenge's **third largest event** by building an asteroid collision simulator in **Godot** showcasing real-world seismic data and meteor impact visualizations.